Electric Forces: Reading

Chemistry · Unit 1B · Level 3 (closer to the class work) · goes with L1.7-L1.9 (Coulomb's Law, Neutral Interactions, Static Interaction Review)

Name: _____

Pushes and Pulls Without Touching

Charged objects push or pull on each other without touching. This push or pull is called the electric force. Opposite charges (+ and −) attract. Like charges (− and −, or + and +) repel. Two balloons rubbed on your hair are both negative, so they push apart.

Charles-Augustin de Coulomb measured electric forces in the 1780s. His rule, Coulomb's law, says two things control the force: the amount of charge and the distance between the objects. More charge makes a stronger force. More distance makes a weaker force.

Distance matters a lot. If you double the distance, the force does not just get cut in half. It becomes one fourth (¼) as strong, because 2 × 2 = 4. If you triple the distance, the force becomes one ninth (1/9), because 3 × 3 = 9. Charge is simpler: double the charge on one object, and the force doubles.

The numbers of electrons are huge. A rubbed balloon can have about 2,000,000,000,000 extra electrons. Scientists write this in scientific notation: 2×10^{12}. The small number (the exponent) tells how many places the decimal moves. A bigger exponent means a much bigger number.

A charged object can even pull on a neutral object. When a negative balloon comes near paper, the paper's electrons are pushed to the far side. The near side becomes slightly positive. Since the positive side is closer to the balloon, the pull is stronger than the push, and the paper jumps up. We say the paper is polarized.

Check your reading

1. Name the two things that control the strength of the electric force.

2. Why does doubling the distance make the force ¼ as strong, not ½?

3. How can a balloon pull on paper that has no charge?